

FBISE PHYSICS
CLASS 10
COMPREHENSIVE GUESS PAPER
2nd Annual Examination 2026

EXAM DATE	PREDICTED PATTERN	THEORY MARKS
24 September 2026	12 MCQs + 11×3 + 4×5	65

Prepared from the official Summative/Formative SLO framework, Class 10 Physics textbook, FBISE Model Paper 2025, 1st Annual papers 2025 and 2026, and 2nd Annual Paper 2025.

Prepared by Fakhar STEM Sphere

High-probability prediction + reserve question bank + formula revision

Important: A guess paper can improve targeted preparation but cannot guarantee exact repetition or full marks. Students should understand and practise every listed Summative concept.

1. Evidence-Based Prediction Map

The strongest prediction is the repeated 2026 slot structure, not blind chapter guessing. A+ means repeated in the current blueprint and/or model-paper long/short structure; A means strong recurring SLO and past-paper support; B is a useful reserve.

Priority	Clusters	Why they matter
A+	Conduction; loudness/pitch; TIR + critical angle; transformer; DC motor; parallel resistance; half-life; electrical energy/cost	Repeated across model paper and/or multiple 2026 variants, often in fixed Section B/C slots.
A	Specific heat; thermal expansion; evaporation; greenhouse effect; waves/diffraction; diode; logic gates; Lenz's law; EM spectrum; lens images; nuclear fission/fusion	Strong Summative alignment with recent recurrence and numerical potential.
B	Electroscope/lightning; ultrasound; dispersion; eye defects; field patterns; resistivity; domestic wiring; carbon dating; alpha scattering	Summative and examinable; valuable as reserve questions when FBISE rotates SLO coverage.

Recommended use: First complete the Primary Predicted Paper without choice. Then cover every A+ and A reserve question. Use the B list only after the first two levels are secure.

Source audit

Documents reviewed: Summative/Formative Topics (24 PDF pages); Class 10 Physics textbook (364 pages); Model Paper 2025 (4 pages); 1st Annual 2025 Hard Area (3 pages); 1st Annual 2025 Local Area (3 pages); 1st Annual 2026 three-paper set (9 pages); and 2nd Annual 2025 (2 pages). Scanned papers were OCR-checked against their page images. Older questions were retained only when their SLO remains Summative for theory.

2. Primary Predicted Paper - Section A

Attempt all 12 MCQs. These 12 form the strongest compact prediction; the remaining 36 MCQs appear later as the reserve bank.

1. When a wave enters shallow water, which quantity remains unchanged?
(A) Speed (B) Frequency (C) Wavelength (D) Amplitude
2. The relation between coefficient of linear expansion α and coefficient of volume expansion β is:
(A) $\beta = 3\alpha$ (B) $\alpha = 3\beta$ (C) $\beta = \alpha/3$ (D) $\beta = \alpha$
3. Heat transferred to a body is calculated by:
(A) $Q = mc\Delta T$ (B) $Q = m/c\Delta T$ (C) $Q = c\Delta T/m$ (D) $Q = m\Delta T/c$
4. Heat from the Sun reaches Earth mainly by:
(A) Conduction (B) Convection (C) Radiation (D) Evaporation
5. Pitch of a sound depends mainly on its:
(A) Amplitude (B) Frequency (C) Speed (D) Intensity
6. Total internal reflection occurs when light travels:
(A) Rarer to denser, $i > c$ (B) Denser to rarer, $i > c$ (C) Denser to rarer, $i < c$ (D) Normally into any medium
7. For a medium in air, refractive index and critical angle are related by:
(A) $n = \sin c$ (B) $n = 1/\sin c$ (C) $n = \cos c$ (D) $n = 1/\cos c$
8. Charging a solid by friction involves transfer of:
(A) Protons (B) Neutrons (C) Electrons (D) Atoms
9. Electric current is given by:
(A) $I = Qt$ (B) $I = Q/t$ (C) $I = t/Q$ (D) $I = VQ$
10. A semiconductor diode is mainly used in a circuit as a:
(A) Transformer (B) Rectifier (C) Capacitor (D) Resistor
11. Force on a current-carrying conductor is maximum when it is:
(A) Parallel to the field (B) At 45° to the field (C) Perpendicular to the field (D) Opposite to the field
12. A transformer operates on the principle of:
(A) Electrostatic induction (B) Mutual induction (C) Thermionic emission (D) Coulomb's law

Primary MCQ key: 1-B | 2-A | 3-A | 4-C | 5-B | 6-B | 7-B | 8-C | 9-B | 10-B | 11-C | 12-B

3. Primary Predicted Paper - Section B

Answer all 11 parts. Prepare both alternatives; FBISE may interchange questions within the repeated conceptual slots. Each part carries 3 marks.

Q2 (1) Define specific heat capacity; write $Q = mc\Delta T$ and its SI unit. Calculate the heat needed to raise 2.0 kg of a substance ($c = 420 \text{ J kg}^{-1} \text{ K}^{-1}$) from 40°C to 45°C .

OR

Define isotope and nuclide. Explain whether isotopes of an element have similar chemical properties, with one example.

Q2 (2) Describe an electric field and state its direction. Draw field-line patterns around $+q$ and $-q$.

OR

Explain forward and reverse biasing of a diode. State its role in rectification.

Q2 (3) Compare transverse and longitudinal waves by particle vibration and energy-transfer directions; give one example of each.

OR

Four identical 2.0 V cells are connected (a) in series and (b) in parallel. Find the total emf in each case.

Q2 (4) A 1.0 kW heater runs for 5 h. Calculate energy in kWh and cost at Rs 30 per unit.

OR

Define refractive index. Calculate the speed of light in a medium of refractive index 1.5.

Q2 (5) State Lenz's law and explain, using a moving magnet and coil, why it agrees with conservation of energy.

OR

An appliance operates at 220 V and draws 2.0 A. Calculate its power and energy used in 5 minutes.

Q2 (6) Differentiate between alternating current and direct current and sketch their current-time graphs.

OR

Explain, using specific heat and convection, why coastal areas have a more moderate climate than inland areas.

Q2 (7) Define echo and derive/calculate the minimum source-obstacle distance for a distinct echo when $v = 340 \text{ m s}^{-1}$ and persistence of hearing is 0.10 s.

OR

Explain why a charged insulator is discharged by (a) passing it over a flame and (b) exposing it to damp air.

Q2 (8) Write the electromagnetic spectrum in increasing frequency (or decreasing wavelength). State the speed of EM waves in vacuum.

OR

State Ohm's law and its condition. A 6Ω resistor is connected across 12 V; calculate current.

Q2 (9) Define half-life. A 160 mg sample has half-life 10 h. Find the amount remaining after 40 h.

OR

An object 5 cm high forms an image 10 cm high. Calculate linear magnification and state when magnification is greater than one.

Q2 (10) Differentiate between real and virtual images. State the characteristics of the image formed by a magnifying glass.

OR

Explain how wavelength and gap width affect diffraction. State when diffraction is maximum.

Q2 (11) Differentiate between evaporation and boiling. State three factors affecting evaporation and explain evaporative cooling.

OR

Explain why the sky is blue by day and reddish at sunrise/sunset.

4. Primary Predicted Paper - Section C

Attempt all four questions. Prepare both alternatives. Each question carries 5 marks; include equations, labeled diagrams, units, and a final numerical answer where required.

Q3. Explain conduction in non-metals by lattice vibrations and in metals by lattice vibrations plus free electrons. Add a labeled particle-level sketch.

OR

Explain nuclear fission and controlled chain reaction in a reactor. Compare fission with fusion and identify the Sun's energy source.

Q4. Explain how amplitude and frequency affect loudness and pitch. Draw comparative waveforms for (a) loud/soft and (b) high/low pitch.

OR

Define total internal reflection, state its two conditions, draw a ray diagram, and derive $n = 1/\sin c$.

Q5. Explain real and apparent expansion of liquids and why apparent expansion is less than real expansion. Include a suitable experimental description.

OR

Derive equivalent resistance for two resistors connected in parallel. Then find the equivalent resistance of two $10\ \Omega$ resistors in parallel.

Q6. Explain the construction, principle, and working of a simple iron-core transformer. Derive/use $V_p/V_s = N_p/N_s$ and distinguish step-up from step-down.

OR

Explain the construction and operation of a DC motor, including the functions of brushes and the split-ring commutator. Draw a labeled diagram.

High-scoring answer rule: begin with the definition/principle, develop the explanation in steps, add the required labeled diagram or derivation, substitute values with SI units, and box the final result.

5. Reserve MCQ Bank - 36 Extra Questions

These complete a 48-MCQ high-yield bank. All are drawn from theory-Summative concepts.

13. At standard atmospheric pressure, water boils at:
(A) 0°C (B) 50°C (C) 100°C (D) 273°C
14. Which change increases the rate of evaporation?
(A) Higher humidity (B) Smaller surface area (C) Lower temperature (D) Faster air movement
15. During melting at the melting point, the temperature of a pure substance:
(A) Increases (B) Decreases (C) Remains constant (D) Becomes 0 K
16. Gases expand more than liquids and solids because their particles are:
(A) More tightly bound (B) Farther apart (C) Motionless (D) Heavier
17. A dull black surface is generally a:
(A) Poor absorber and emitter (B) Good reflector only (C) Good absorber and emitter (D) Perfect insulator
18. A greenhouse gas warms Earth mainly by absorbing:
(A) Ultraviolet radiation (B) Infrared radiation (C) Radio waves (D) Sound waves
19. Wave speed is:
(A) $v = f\lambda$ (B) $v = f/\lambda$ (C) $v = \lambda/f$ (D) $v = f + \lambda$
20. Sound cannot travel through:
(A) Steel (B) Water (C) Air (D) Vacuum
21. The normal audible range for a human is approximately:
(A) 2-200 Hz (B) 20-20,000 Hz (C) 200-200,000 Hz (D) Above 20,000 Hz only
22. Sound above 20 kHz is called:
(A) Infrasound (B) Ultrasound (C) Audible sound (D) Noise
23. Diffraction is greatest when gap size is:
(A) Much larger than wavelength (B) Comparable to wavelength (C) Zero (D) Independent of wavelength
24. An echo is produced by:
(A) Refraction of sound (B) Reflection of sound (C) Absorption of sound (D) Interference only
25. Light bends towards the normal when it enters a medium in which it:
(A) Speeds up (B) Slows down (C) Keeps the same speed (D) Stops
26. A converging lens used as a magnifying glass forms an image that is:
(A) Real, inverted, diminished (B) Virtual, upright, magnified (C) Real, upright, same size (D) Virtual, inverted, diminished
27. Short-sightedness is corrected using a:
(A) Converging lens (B) Diverging lens (C) Plane glass plate (D) Prism
28. The sky appears blue mainly because shorter visible wavelengths are:
(A) Absorbed completely (B) Scattered more (C) Refracted less (D) Reflected by oceans
29. Like electric charges:
(A) Attract (B) Repel (C) Neutralize instantly (D) Have no force
30. The direction of an electric field is the direction of force on a:
(A) Negative test charge (B) Positive test charge (C) Neutron (D) Current element
31. If 10 C passes a point in 5 s, the current is:
(A) 0.5 A (B) 2 A (C) 15 A (D) 50 A
32. For an ohmic conductor at constant temperature:
(A) $V \propto I$ (B) $V \propto 1/I$ (C) $R \propto I$ (D) I is always zero
33. Two resistors in parallel have equivalent resistance:
(A) Greater than both (B) Equal to their sum (C) Less than the smallest (D) Equal to the largest

34. Electrical power is calculated by:
 (A) $P = VI$ (B) $P = V/I$ (C) $P = It$ (D) $P = QVt$
35. One unit of electrical energy equals:
 (A) 1 W (B) 1 kW (C) 1 kWh (D) 1 J
36. Domestic appliances are connected in parallel so each receives:
 (A) The same current (B) The full supply voltage (C) Zero resistance (D) Half the voltage
37. An LED conducts and emits light mainly when it is:
 (A) Reverse biased (B) Forward biased (C) Unbiased (D) Disconnected
38. The output of an OR gate is LOW only when:
 (A) Both inputs are LOW (B) Both inputs are HIGH (C) One input is HIGH (D) Any input is HIGH
39. The output of a NAND gate for inputs 1 and 1 is:
 (A) 0 (B) 1 (C) Undefined (D) Alternating
40. The split-ring commutator in a DC motor:
 (A) Stores charge (B) Reverses coil current every half-turn (C) Increases resistance (D) Produces AC supply
41. Increasing the number of turns in a generator coil generally:
 (A) Decreases induced emf (B) Increases induced emf (C) Makes emf zero (D) Changes AC to DC
42. Lenz's law is consistent with conservation of:
 (A) Charge only (B) Momentum only (C) Energy (D) Mass only
43. All electromagnetic waves in vacuum travel at approximately:
 (A) 3×10^8 m/s (B) 340 m/s (C) 3×10^6 m/s (D) 9.8 m/s
44. An isotope has the same proton number but a different:
 (A) Electron charge (B) Nucleon number (C) Chemical symbol (D) Proton number
45. The greatest ionizing power is possessed by:
 (A) Alpha particles (B) Beta particles (C) Gamma rays (D) Visible light
46. The greatest penetrating power is possessed by:
 (A) Alpha particles (B) Beta particles (C) Gamma rays (D) Protons
47. After three half-lives, the undecayed fraction is:
 (A) $1/2$ (B) $1/4$ (C) $1/8$ (D) $1/16$
48. The energy released by mass conversion is calculated from:
 (A) $E = mc$ (B) $E = mc^2$ (C) $E = m/c^2$ (D) $E = c^2/m$

Reserve MCQ key

13: C	14: D	15: C	16: B	17: C	18: B	19: A	20: D	21: B
22: B	23: B	24: B	25: B	26: B	27: B	28: B	29: B	30: B
31: B	32: A	33: C	34: A	35: C	36: B	37: B	38: A	39: A
40: B	41: B	42: C	43: A	44: B	45: A	46: C	47: C	48: B

6. Extra Short-Question Bank

Use this bank after the 22 primary alternatives. Questions are grouped by domain so students can close chapter-level gaps.

Heat and Thermodynamics

1. Explain thermal expansion using kinetic theory and compare expansion in solids, liquids, and gases.
2. Why are expansion gaps left in railway tracks and bridges? Why are overhead wires kept slightly loose?
3. Distinguish melting, solidification, boiling, condensation, sublimation, and deposition in terms of energy transfer.
4. Explain why temperature stays constant during a change of state and define specific latent heat.
5. Explain convection in liquids and gases in terms of density changes; describe one demonstration.
6. Explain how convection in seawater supports marine life and how thermals help birds/gliders.
7. Compare dull black and polished surfaces as absorbers, emitters, and reflectors of infrared radiation.
8. Explain greenhouse effect, global warming, and their link to extreme weather/cyclones.

Waves, Sound and Optics

9. Define wavelength, frequency, time period, amplitude, crest, trough, compression, rarefaction, and wavefront.
10. Show that waves transfer energy without transferring matter.
11. Explain how a tsunami changes as it enters shallow water.
12. Explain why sound travels faster in solids than liquids and gases, and why it cannot travel in vacuum.
13. Define intensity, intensity level, pitch, loudness, and timbre/quality of sound.
14. State the uses of ultrasound in medical scanning, cleaning, and sonar; include a distance/depth calculation.
15. Explain noise pollution, its effects, acoustic protection, and the use of soft materials in halls/classrooms.
16. Apply laws of reflection; describe the image formed by a plane mirror.
17. Apply Snell's law $n_1 \sin i = n_2 \sin r$ in a numerical problem.
18. Explain optical-fibre communication by total internal reflection and give its advantages.
19. Draw ray diagrams for a converging lens for object positions beyond $2F$, at $2F$, between F and $2F$, and inside F .
20. Draw the normal, myopic, and hypermetropic eyes and show their correction using suitable lenses.
21. Explain dispersion through a prism and arrange VIBGYOR by frequency/wavelength.

Electricity, Electronics and Electromagnetism

22. Explain charging by friction and charging a conductor by induction followed by earthing.
23. Explain lightning formation, electrical breakdown, corona discharge, and protection by a lightning rod.
24. Differentiate e.m.f. from potential difference and apply $E = W/Q$ or $V = W/Q$ in a numerical.
25. Differentiate conductors and insulators using the electron model.
26. Differentiate actual electron flow from conventional current.
27. Explain resistance and resistivity; state factors affecting resistance and interpret V-I graphs.
28. Derive equivalent resistance for a series combination and compare series with parallel connections.

29. Explain the functions of live, neutral, and earth wires and why appliance casings are earthed/double insulated.
30. Draw symbols and truth tables for AND, OR, NAND, NOR, and NOT gates.
31. Explain a burglar alarm or fire alarm using logic-gate action.
32. Describe electromagnetic induction and factors affecting induced emf.
33. Explain an AC generator with coil positions corresponding to maximum, zero, and reversed emf.
34. Draw magnetic fields around a straight wire and solenoid; state effects of reversing/increasing current.
35. Explain why high-voltage transmission reduces I^2R power loss.

Modern Physics

36. State Rutherford alpha-scattering findings and the conclusions about the nucleus.
37. Use nuclide notation to find protons, neutrons, and electrons.
38. Compare alpha, beta, and gamma radiations by nature, charge, mass, ionization, penetration, and deflection.
39. Complete alpha-, beta-, and gamma-decay equations using conservation of nucleon and proton numbers.
40. Explain spontaneous and random radioactive decay and define activity.
41. Interpret a decay curve and calculate number/mass/activity after several half-lives.
42. Explain carbon dating and solve a simple age problem.
43. Apply $E = mc^2$ to a nuclear energy calculation.

7. Extra Long Questions and Numericals

These are the most useful reserve problems for 5-mark preparation. Students should show formula, substitution, SI unit, working, and final statement.

1. Specific heat numerical: A 0.50 kg metal absorbs 9,000 J and its temperature rises by 40 K. Find its specific heat capacity.
2. Volume expansion numerical: A cube of side 5.0 cm at 15°C is heated to 100°C; $\beta = 9 \times 10^{-6} \text{ K}^{-1}$. Find initial volume, increase, and final volume.
3. Wave numerical: 200 waves pass a point in 40 s; $\lambda = 4.0 \text{ cm}$. Find frequency, period, and wave speed.
4. Echo/SONAR numerical: An echo returns after 0.80 s in water where $v = 1500 \text{ m s}^{-1}$. Find depth.
5. Refraction numerical: Light enters glass ($n = 1.50$) from air at $i = 30^\circ$. Find r using Snell's law.
6. Critical-angle numerical: Find the critical angle for glass of refractive index 1.50.
7. Lens numerical: An object 2 cm high is placed 30 cm from a converging lens and forms an image 6 cm high. Find magnification and image distance using $m = h_i/h_o = v_i/u_o$.
8. Potential-difference numerical: Find the work done in moving 10 C of charge through a potential difference of 100 V.
9. Current numerical: Calculate the number of electrons crossing a section in 16 s when $I = 1.0 \text{ A}$ ($e = 1.6 \times 10^{-19} \text{ C}$).
10. Resistance numerical: A wire has $\rho = 1.7 \times 10^{-8} \Omega \text{ m}$, length 2.0 m, and area $1.0 \times 10^{-6} \text{ m}^2$. Find R .
11. Parallel network numerical: 5 Ω , 10 Ω , and 15 Ω are connected across 30 V. Find equivalent resistance, total current, and branch currents.
12. Electrical cost numerical: Four 100 W bulbs operate 6 h daily for 30 days. Find energy and cost at Rs 40 per kWh.
13. Transformer numerical: A step-down transformer changes 240 V to 12 V. If $N_p = 2000$, find N_s ; state the current relation for an ideal transformer.
14. Induction application: Predict induced-current direction as a north pole approaches/leaves a coil and justify with Lenz's law.
15. Half-life numerical: A 300 g sample has a half-life of 6 h. Find the amount after 18 h and the decayed fraction.
16. Mass-energy numerical: Calculate energy released when 1.0 mg of mass is converted completely to energy.

8. Rapid Formula and Diagram Checklist

Topic	Must-know relations
Heat	$Q = mc\Delta T$; $\Delta V = \beta V_0 \Delta T$; $\beta = 3\alpha$
Waves/Sound	$v = f\lambda$; $T = 1/f$; echo/SONAR distance = $vt/2$
Refraction/Lens	$n = c/v$; $n_1 \sin i = n_2 \sin r$; $n = 1/\sin c$; $m = h_i/h_o = v_i/u_o$; $P = 1/f(m)$
Electrostatics/e.m.f.	$E(\text{field}) = F/q$; e.m.f. = W/Q ; potential difference $V = W/Q$
Current/Resistance	$I = Q/t$; $V = IR$; $R = \rho L/A$; $R_s = R_1 + R_2$; $1/R_p = 1/R_1 + 1/R_2$
Power/Energy	$P = VI = I^2 R = V^2/R$; $W = Pt = VIt$; cost = kWh $\times$ tariff
Transformer/Induction	$V_p/V_s = N_p/N_s$; ideal $V_{plp} = V_{sls}$; induced emf increases with turns and rate of flux change
Nuclear	$N = N_0(1/2)^n$ where $n=t/T_{1/2}$; $E = mc^2$

Diagrams students must practise

Conduction in metallic/non-metallic solids • Convection current/land and sea breeze • Comparative sound waveforms • Diffraction at narrow/wide gaps • TIR and optical fibre • Converging-lens ray diagrams and eye correction • Electric field patterns/electroscope • Series and parallel resistor circuits • Diode forward/reverse bias and LED • Five logic-gate symbols/truth tables • AC generator and emf-time graph • DC motor with brushes/commutator • Transformer • Controlled nuclear fission/chain reaction • Radioactive decay curve

Unit traps

Convert cm to m, mm to m, μC to 10^{-6} C , mg to kg, minutes/hours to seconds when using SI equations, and watts to kilowatts for electricity bills. Frequency does not change at a boundary; wave speed and wavelength do.

9. Explicitly Excluded from Theory Prediction

The following were omitted because the uploaded assessment framework labels them Formative or PBA-only. This is especially important because some older papers contain legacy questions that no longer belong to the theory-Summative list.

1. Superconductivity near absolute zero (P-10-C-28, C-29)
2. Geothermal activity explanation (P-10-F-25)
3. Conversion of sound by the eardrum/nerves into electrical signals (P-10-D-31)
4. Camera, projector, and photographic-enlarger applications under P-10-D-53 (magnifying glass itself remains Summative under D-50)
5. Biological color vision, gravitational lensing, and acoustic lenses (P-10-D-56 to D-58); correction of short/long sight under D-54 and D-55 remains Summative
6. Atmospheric-lightning types still under research (P-10-E-19)
7. Relay switching, transistor as a switch, circuit memory and quantum-computer overview (P-10-E-47, E-68 to E-72)
8. Deflection of an electron beam and oscilloscope waveform interpretation (P-10-E-92, E-93)
9. Applications and biological damage of electromagnetic waves (P-10-E-96, E-97); spectrum order and sky-color scattering remain Summative
10. Background radiation sources and radiation safety/application-selection content (P-10-F-06, F-07, F-19 to F-21)
11. PBA-only experimental SLOs, including direct laboratory determination of specific/latent heat and block-refraction practicals, unless a theory-Summative statement separately covers the concept
12. Legacy questions not present in the uploaded theory-Summative list, such as Coulomb-law numericals, lens power, telescope/compound microscope, fax/storage-device/ICT questions, and CRO construction

Boundary note: where one subtopic is Formative but a nearby concept is separately marked Summative, only the Summative concept has been retained. For example, magnifying glass is retained, while broader optical-device applications under the Formative SLO are not.

Final 3-Day Priority

Day 1	Conduction + expansion + evaporation; waves/sound; TIR/lenses; solve all related numericals.
Day 2	Ohm/resistor networks + electrical energy; diode/gates; induction, motor, transformer.
Day 3	Half-life + fission/fusion + radiation; complete the primary paper; revise formulae, units, and diagrams.

Prepared by Fakhar STEM Sphere

Study smart, write scientifically, and never omit units or labeled diagrams.